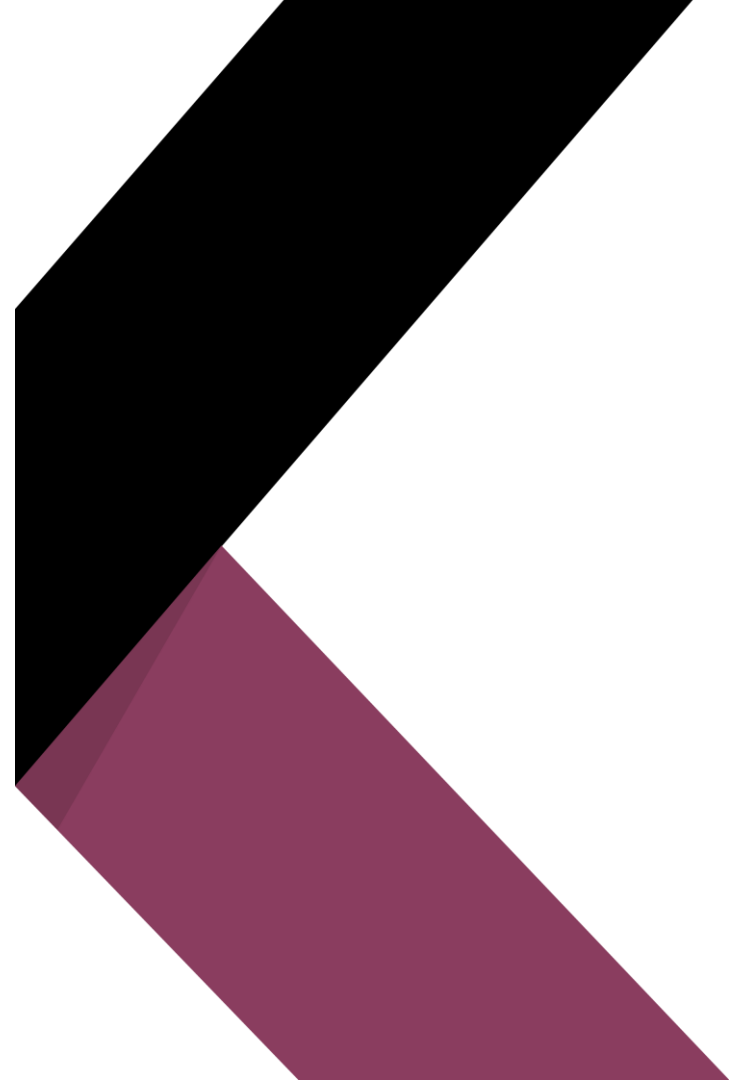


Development
Process

V-Cycle
Model





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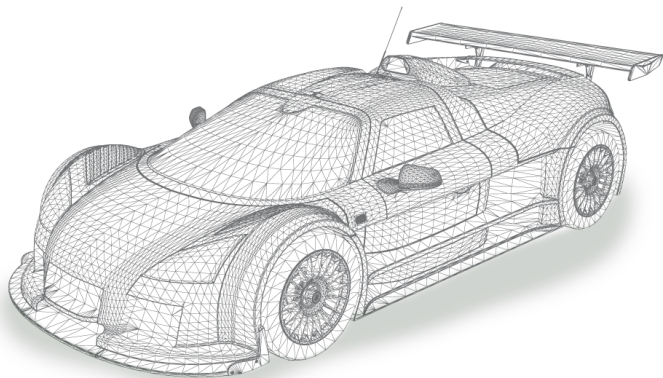
- ❑ **Introduction** to Automotive framework
- ❑ Definition of **Process**
- ❑ The **V-Cycle Process** and its characteristics
- ❑ An **example** of a running V-Cycle Process



Introduction



What is a Car?



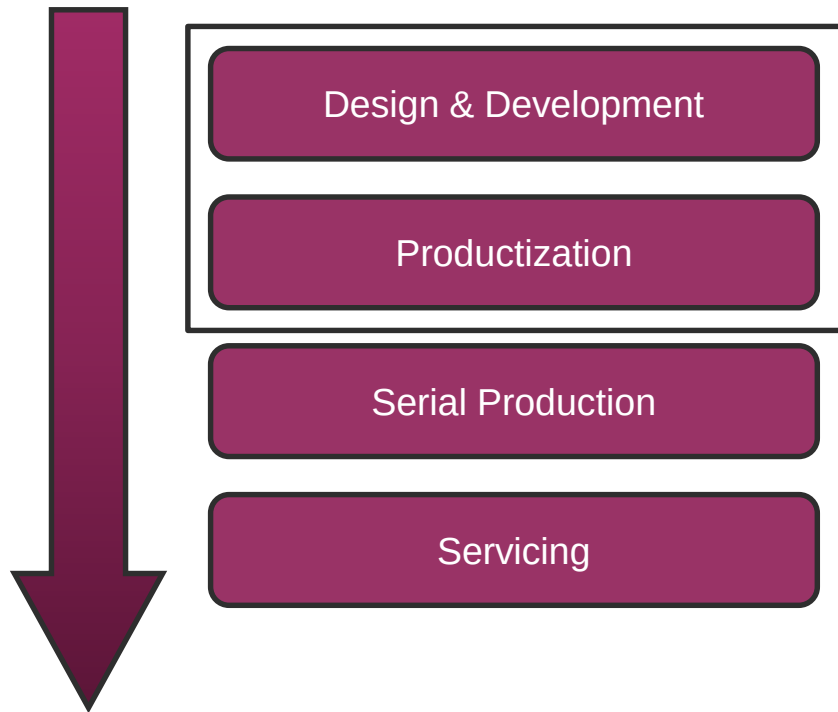
- ❑ Purpose-built ***layers of components***
- ❑ Components:
 - Mechanical
 - Hardware
 - Software
 - Others (Liquids, Oils, Glues...)

The ***complexity*** of assembling and make a product out of these multiple layers is ***high***

An appropriate ***organizational*** effort is required in order to manage the complexity

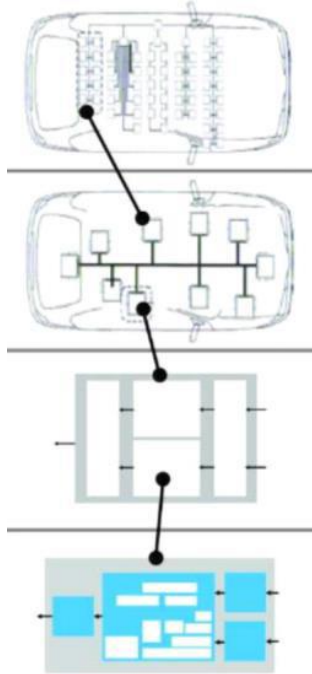
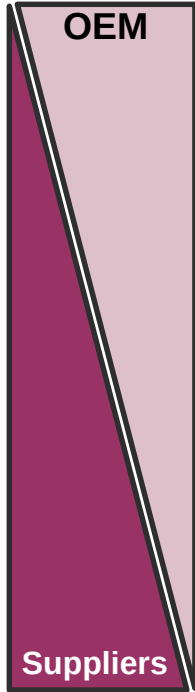


How do we build Car?





Who builds a Car?



DOMAIN	OWNER	DESCRIPTION
Vehicle	OEM	«Original Equipment Manufacturer» i.e. the car-maker (FCA, Ferrari, Audi...)
Subsystem	OEM/Tier 1	Tier 1: Specialized supplier that provides equipment directly to the OEM.
ECU	Tier 1	Tier 1: Specialized supplier that provides equipment directly to the OEM.
Parts (i.e. MCU)	Tier 2	Tier 2: Specialized supplier that provides parts/components to Tier 1



What is a Process?



What is a Process?

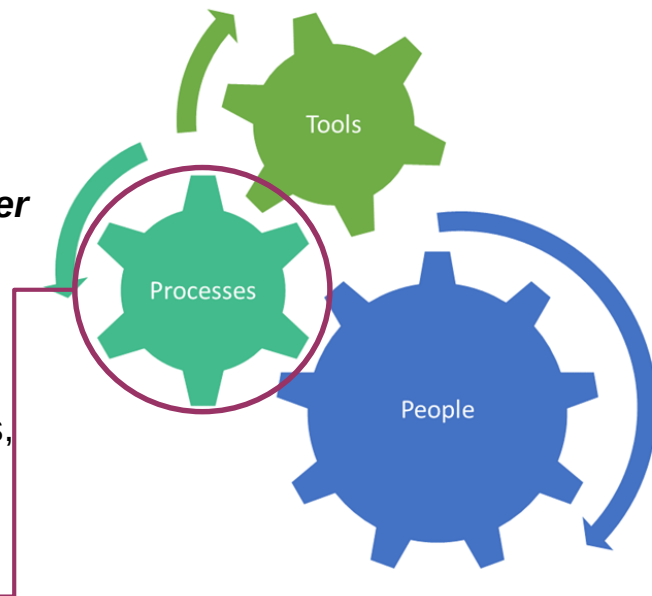
One key concept of success in organizing and improve the operational efficiency of a business is the **People, Process, Tools** framework:

A Process is a **series of steps and actions taken in order** to produce a particular «goal».

It articulates the «**How**» aspect of an organization

It is important that People have **clear roles** in the Process, and that they fully understand it

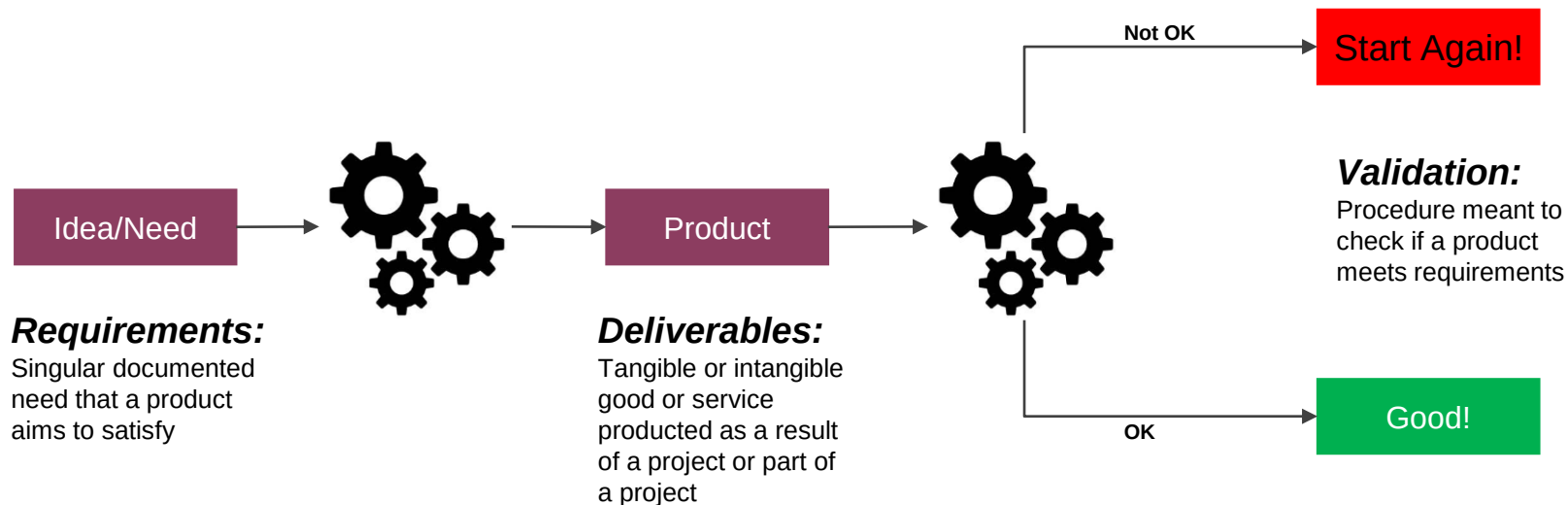
The Tools can improve the Process





What is a Process?

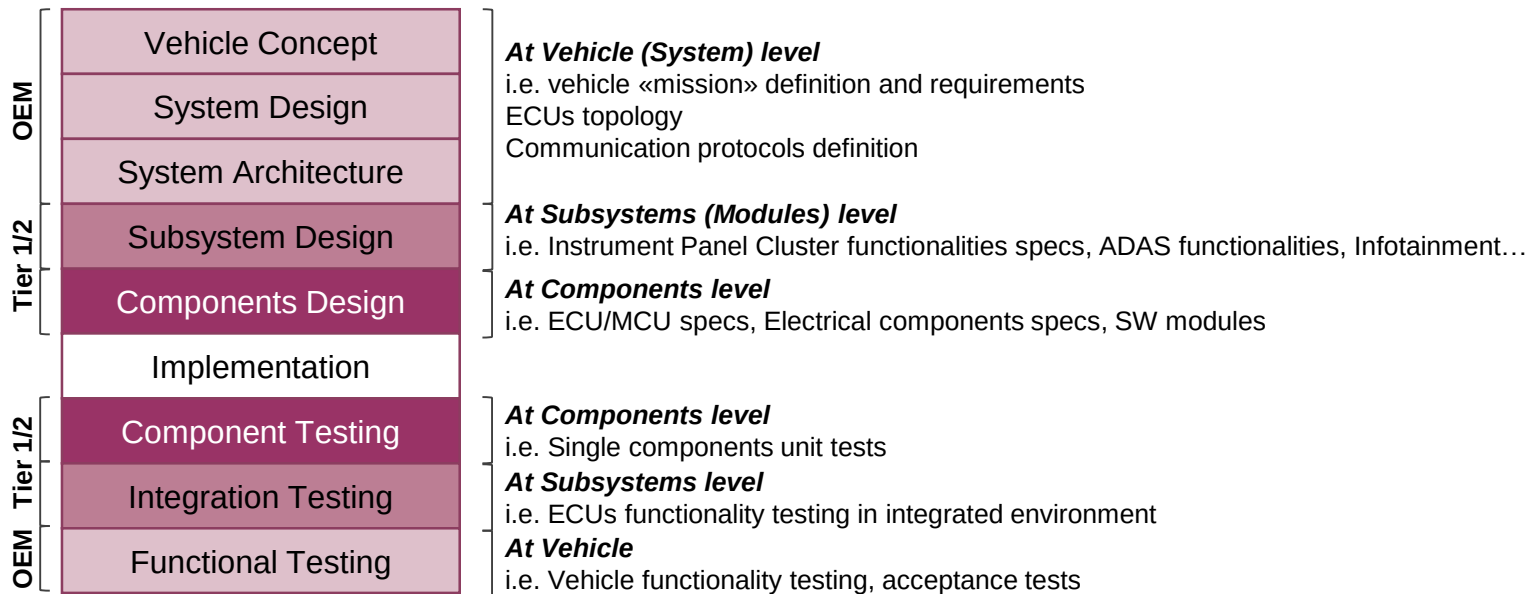
The simplest Process schematization can be this one:





Process for Automotive: System of Systems

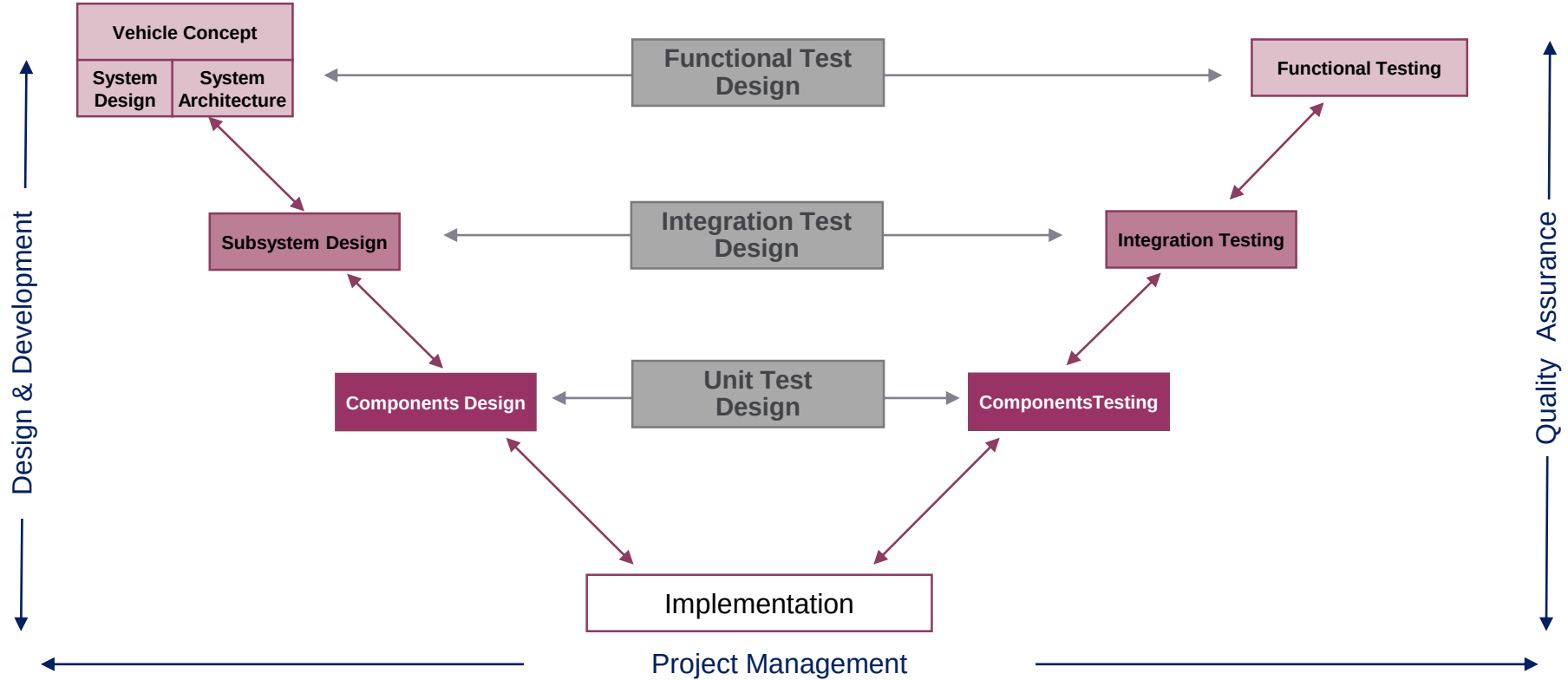
Let's see the steps for a more complex process, as a vehicle development project:



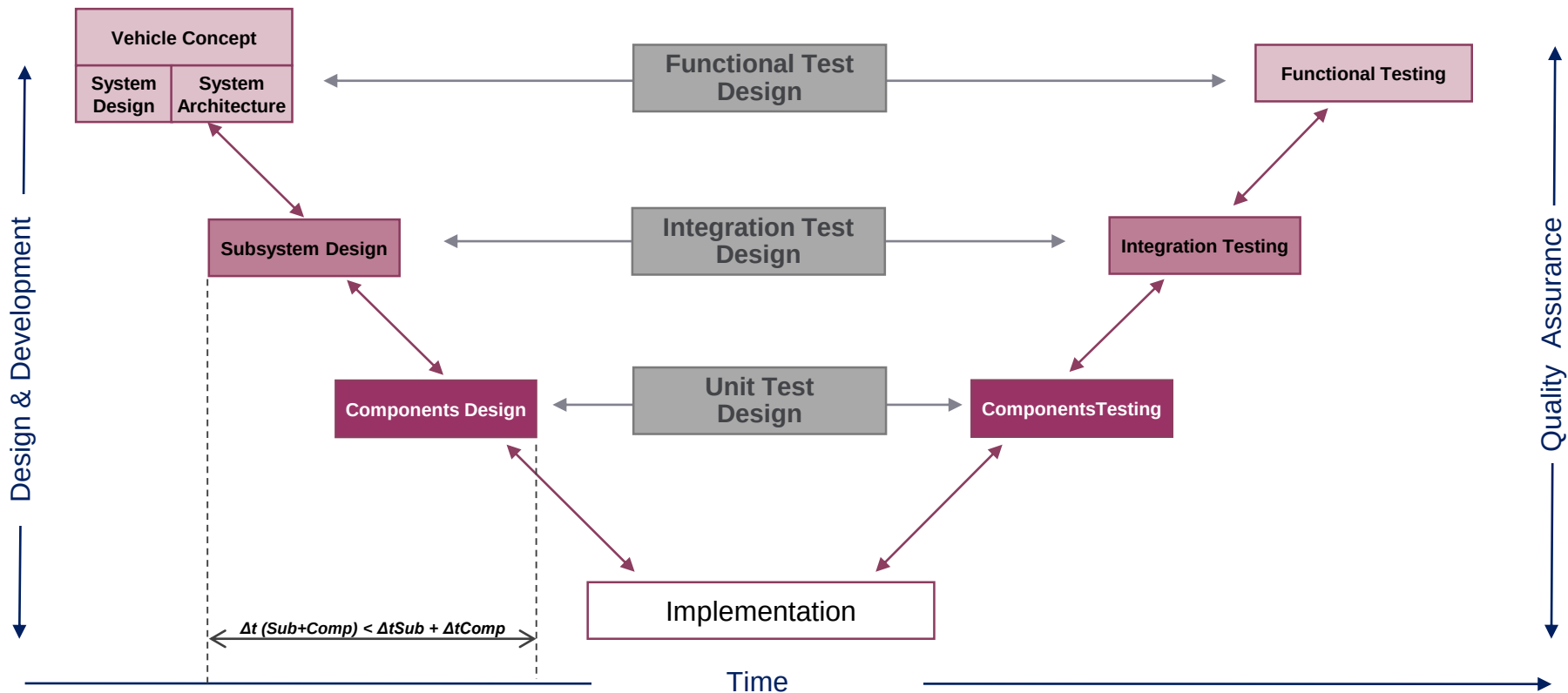


V-Cycle Model

Automotive V-cycle: connecting the branches

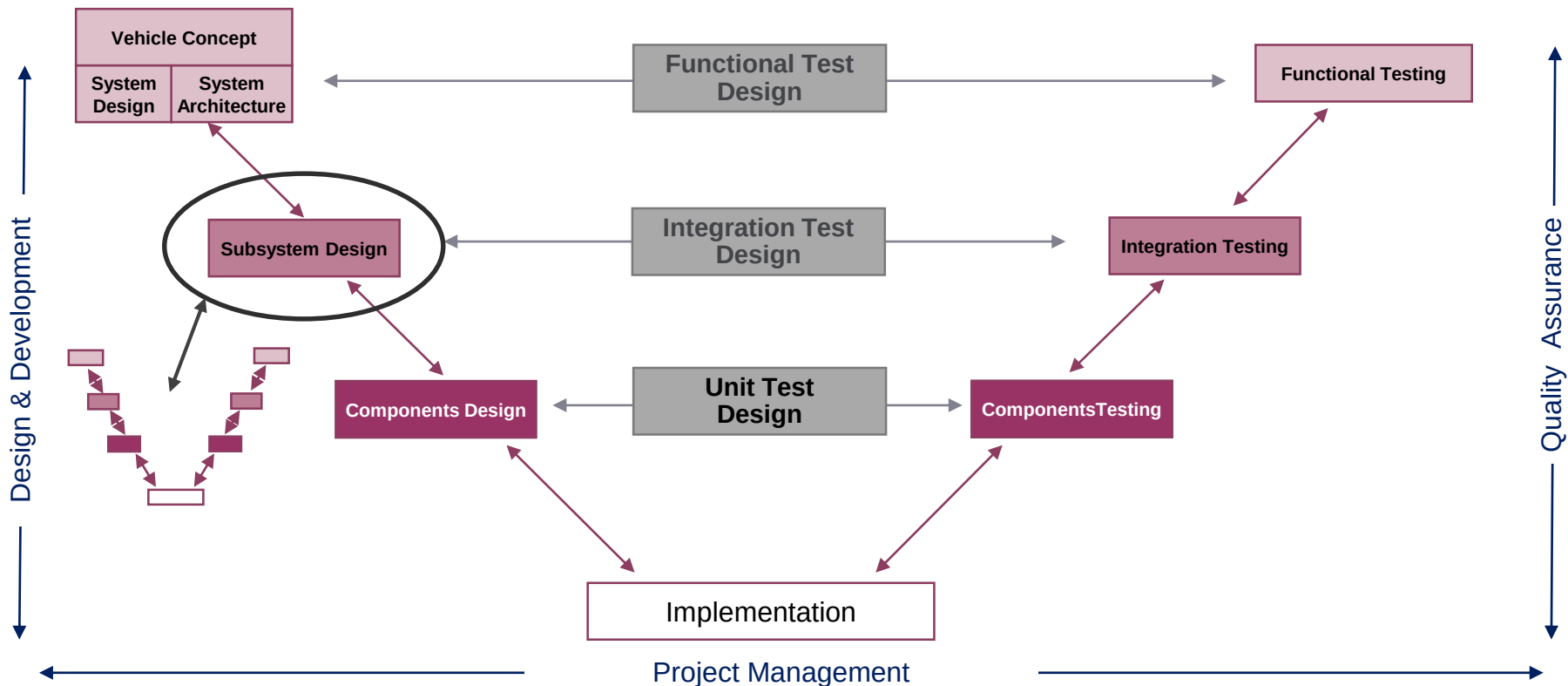


Automotive V-cycle: parallelization of activities



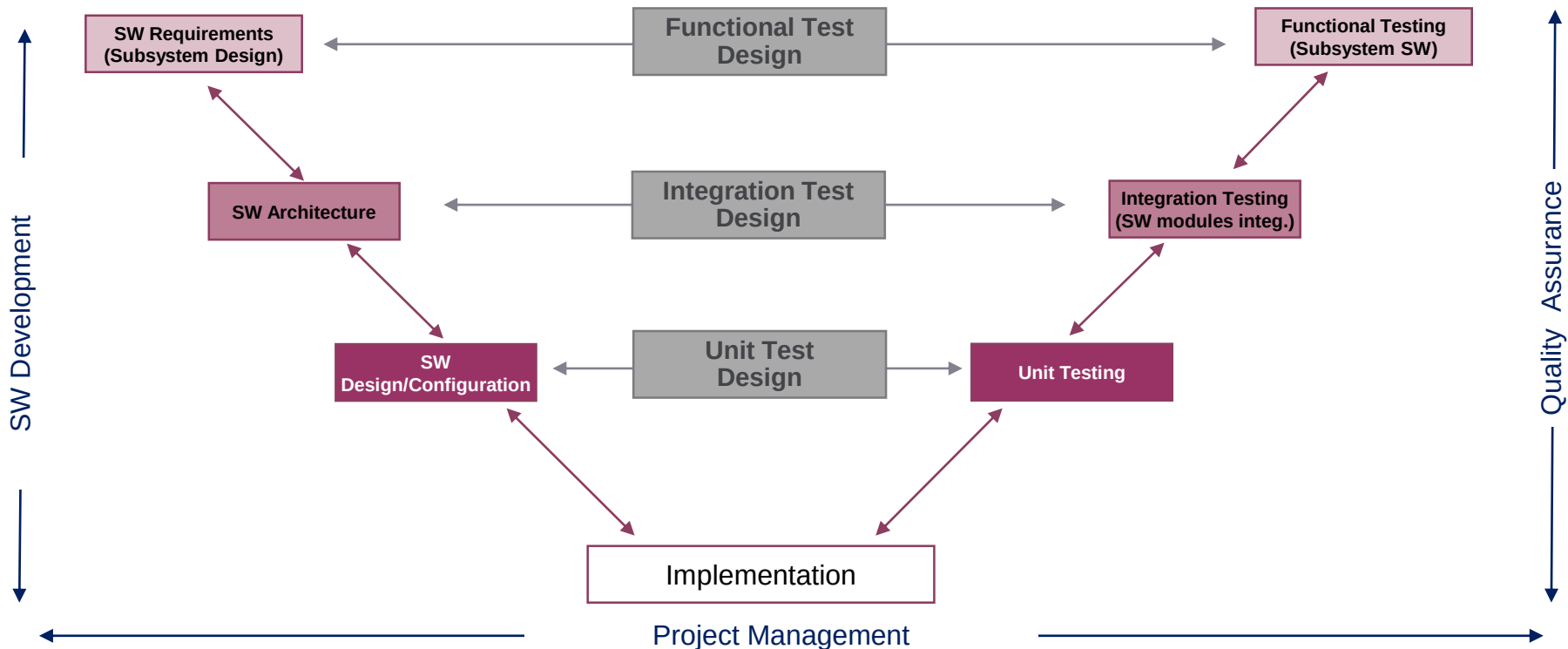


Automotive V-cycle: scalability

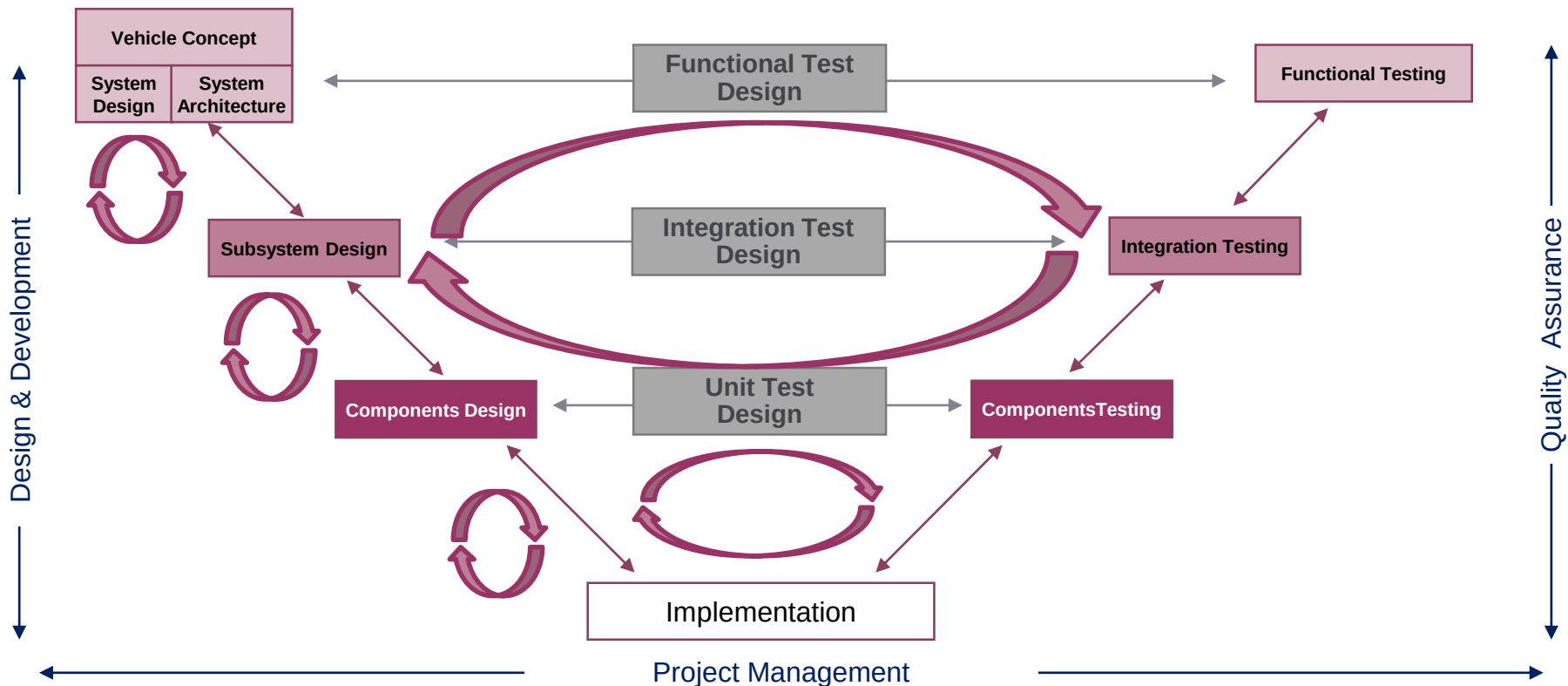




Automotive V-cycle: scalability



Automotive V-cycle: iterative loops and Agility





An example of V-Cycle



An example of a running V-cycle process

Input: n/a
Output: product
requirement/constraints,
functional test design



The **Airbag system** is a critical module of a vehicle: it has to deploy flawlessly in case of a major accident, it doesn't have to deploy when not required, and its **malfunction** shall be promptly and properly **notified** to the final user.

The candidate module to show the Airbag Malfunction signal is the **IPC** (Instrument Panel Cluster).

Functional Safety analysis suggests that the mechanism providing the malfunction information shall be reliable and fail-proof to a certain degree.



Homologation constraints allows the usage of a standard icon, which can be orange or red. Red has been chosen.



An example of a running V-cycle process

Vehicle Concept

Input: n/a

Output: product
requirement/constraints,
functional test design

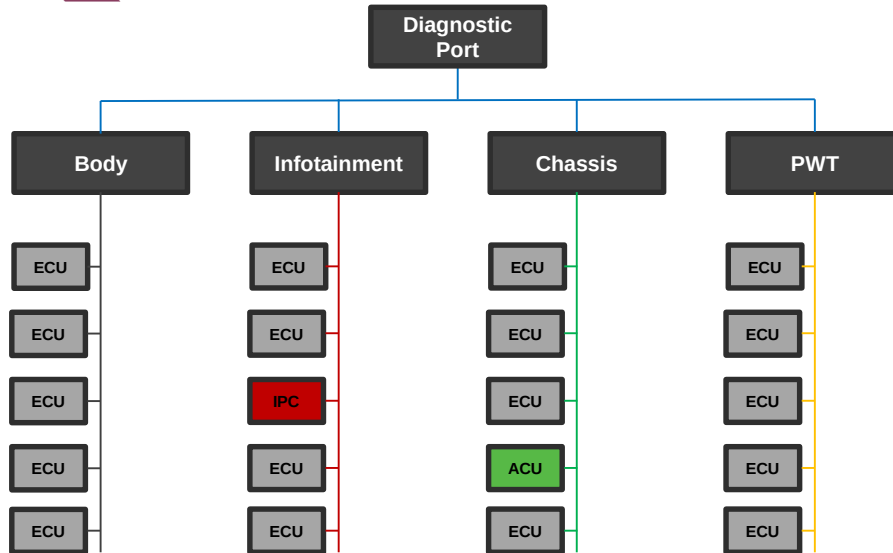


Typical Deliverables of the Vehicle Concept

- **Specification of Requirements** document (list of desired functionalities, definition of vehicle mission, homologation constraints, vehicle safety concept)



An example of a running V-cycle process

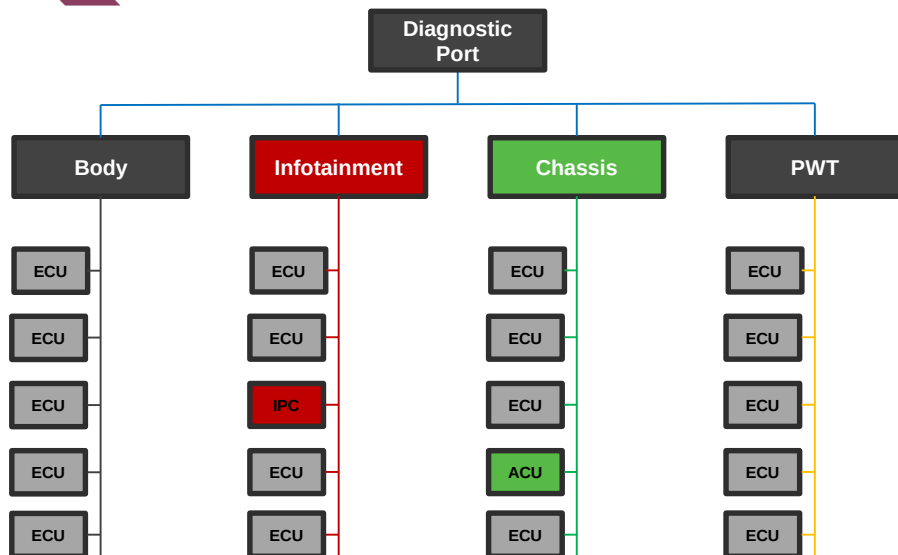


The vehicle will have a domain-based topology:
ECUs are on different communication buses based
on their functionality.

The IPC and the ACU are in two different domains.



An example of a running V-cycle process



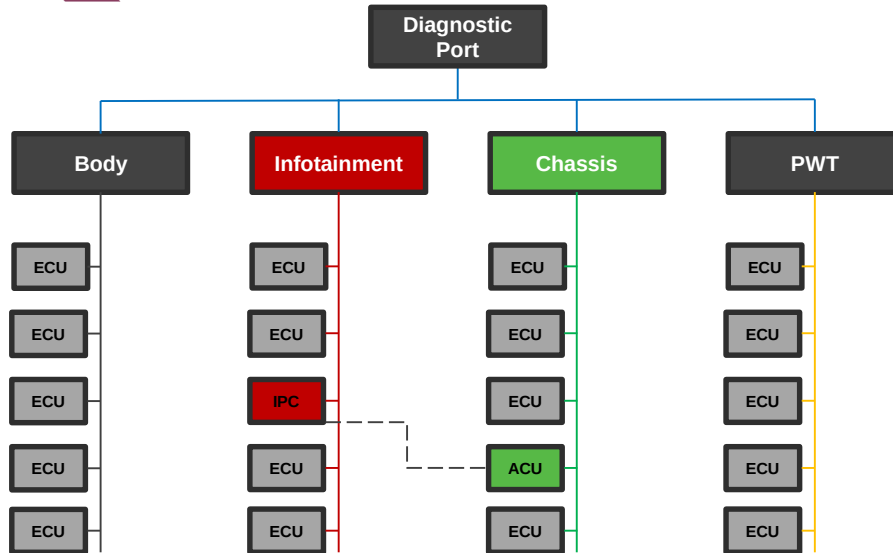
The vehicle will have a domain-based topology: ECUs are on different communication buses based on their functionality.

The IPC and the ACU are in two different domains.

Requirements will be produced to accomplish the desired behaviour: ACU will signal via communication bus to Chassis Domain Controller a malfunction, that will communicate the information to the Infotainment domain and then forwarded to IPC. IPC shall then set the warning lamp ON.



An example of a running V-cycle process



The vehicle will have a domain-based topology: ECUs are on different communication buses based on their functionality.

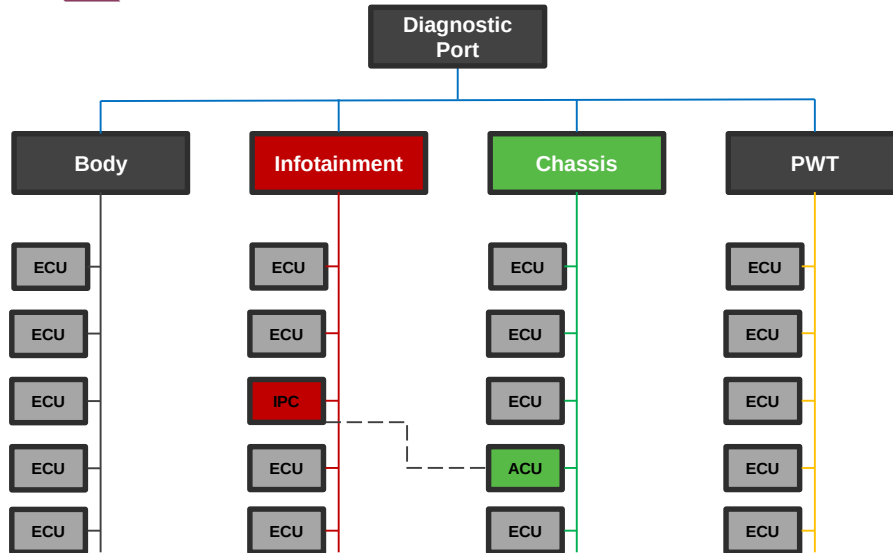
The IPC and the ACU are in two different domains.

Requirements will be produced to accomplish the desired behaviour: ACU will signal via communication bus to Chassis Domain Controller a malfunction, that will communicate the information to the Infotainment domain and then forwarded to IPC. IPC shall then set the warning lamp ON.

An hardwired connection between IPC and ACU is added, to improve the reliability of the functionality as required.



An example of a running V-cycle process



Typical Deliverables of the System Design & Arch.

- ECUs topology
- Communication Bus specs (i.e. CAN Matrix)
- **Vehicle Functions** specifications (i.e. Airbag Management)

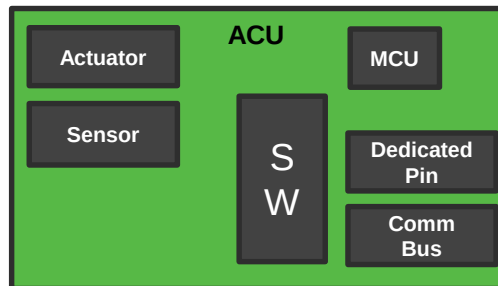


An example of a running V-cycle process

Subsystem Design

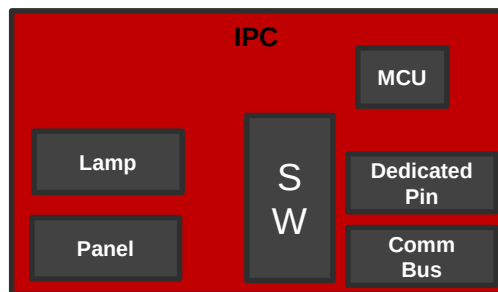
Input: System Design

Output: ECUs I/O and
HW/SW specs,
Integration Test design



ACU shall be capable of diagnostic check on the functionality (sensors, actuators, SW, HW, Communication).

It shall foresee a dedicated signal on communication Bus for malfunction, and shall manage a dedicated hardwired signal accordingly to system status.



IPC shall manage signal on comm Bus and hardwired signal coherently with ACU, and shall foresee a dedicated red lamp and a serigraph on the panel. The lamp shall be turned on upon malfunction signal receiving.



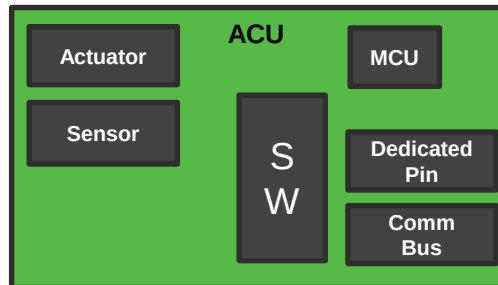
Infotainment and Chassis Domain Controllers shall manage the transfer of malfunction signal.



An example of a running V-cycle process

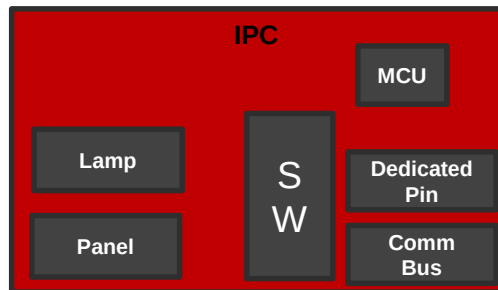
Subsystem Design

Input: System Design
Output: ECUs I/O and
HW/SW specs,
Integration Test design



Typical Deliverables of the Subsystem Design

- Hardware schematics
- **SDD & SRS** (SW Design & Software Requirements Specifications)
- I/O specifications



Infotainment

Chassis



An example of a running V-cycle process

Typical Deliverables of the Components

- Component Datasheet

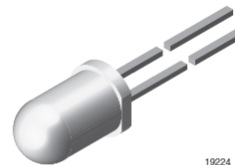
PARTS TABLE														
PART	COLOR	LUMINOUS INTENSITY (mcd)			at If (mA)	WAVELENGTH (nm)			at If (mA)	FORWARD VOLTAGE (V)			at If (mA)	TECHNOLOGY
		MIN.	TYP.	MAX.		MIN.	TYP.	MAX.		MIN.	TYP.	MAX.		
TLUR6400	Red	4	15	-	10	-	630	-	10	-	2	3	20	GaAsP on GaAs
TLUR6401	Red	4	15	32	10	-	630	-	10	-	2	3	20	GaAsP on GaAs

ABSOLUTE MAXIMUM RATINGS (T _{amb} = 25 °C, unless otherwise specified)				
TLUR6401				
PARAMETER	TEST CONDITION	SYMBOL	VALUE	UNIT
Reverse voltage		V _R	6	V
DC forward current		I _F	20	mA
Surge forward current	t _p ≤ 10 μs	I _{FSM}	1	A
Power dissipation	T _{amb} ≤ 65 °C	P _V	60	mW
Junction temperature		T _J	100	°C
Operating temperature range		T _{amb}	- 40 to + 100	°C
Storage temperature range		T _{stg}	- 55 to + 100	°C
Soldering temperature	t ≤ 5 s, 2 mm from body	T _{sd}	260	°C
Thermal resistance junction/ambient		R _{thJA}	500	K/W

OPTICAL AND ELECTRICAL CHARACTERISTICS (T _{amb} = 25 °C, unless otherwise specified)							
TLUR640., RED							
PARAMETER	TEST CONDITION	PART	MIN.	TYP.	MAX.	UNIT	MIN.
Luminous intensity ⁽¹⁾	I _F = 10 mA	TLUR6400	I _V	4	15	-	mcd
		TLUR6401	I _V	4	15	32	mcd
Dominant wavelength	I _F = 10 mA		λ _d	-	630	-	nm
Peak wavelength	I _F = 10 mA		λ _p	-	640	-	nm
Angle of half intensity	I _F = 10 mA		φ	-	± 30	-	deg
Forward voltage	I _F = 20 mA		V _F	-	2	3	V
Reverse voltage	I _R = 10 μA		V _R	6	15	-	V
Junction capacitance	V _R = 0 V, f = 1 MHz		C _j	-	50	-	pF



Universal LED in Ø 5 mm Tinted Diffused Package



19224

DUCT GROUP AND PACKAGE DATA

uct group: LED
age: 5 mm
uct series: standard
e of half intensity: ± 30°

FEATURES

- For DC and pulse operation
- Luminous intensity categorized
- Standard T-1 1/4 package
- TLUR640, without stand-offs
- Material categorization:
For definitions of compliance please see www.vishay.com/doc/999912

APPLICATIONS

- General indicating and lighting purposes

TYPICAL CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)

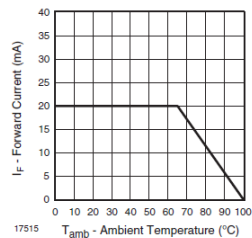


Fig. 1 - Forward Current vs. Ambient Temperature

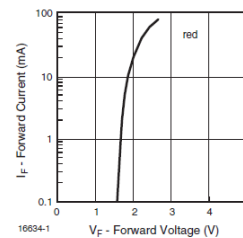


Fig. 4 - Forward Current vs. Forward Voltage



An example of a running V-cycle process

Implementation





An example of a running V-cycle process



Does the single LED turn on when powered?
(Do the components work as they are intended to do?)

Typical «unit tests» are, as example:

- Testing SW modules/portion
- Testing single HW components (MCU, Cap...)
- Testing electrical harness

Typical Deliverables of Component Testing

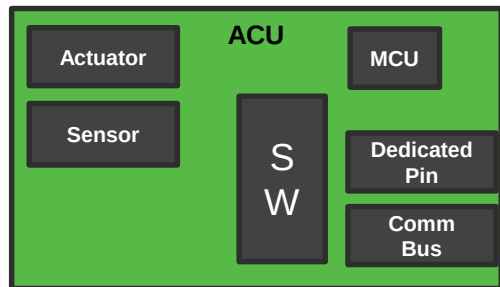
- Component test results document



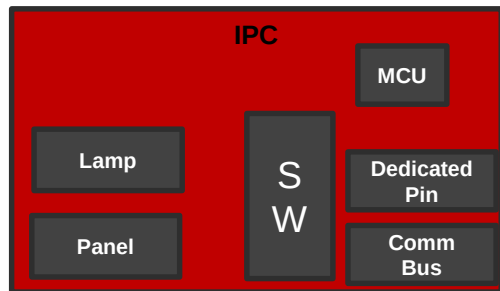
An example of a running V-cycle process

Integration Tests

Input: Integration Test Design, Unit Test Results
Output: Integration Test Result



Does ACU diagnosis malfunction of the airbag deployment system, and trigger properly the signal on comm bus and hardwire?



Does the IPC process incoming malfunction signals on Bus and hardwire, and turn on the LED?

Infotainment

Chassis

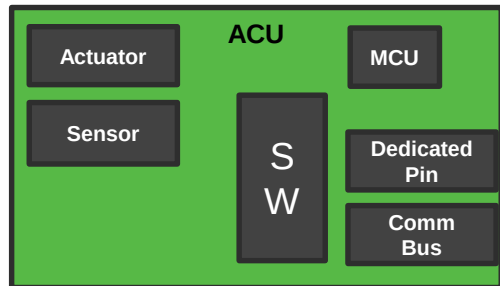
Are the Domain Controllers able to gateway the signal?



An example of a running V-cycle process

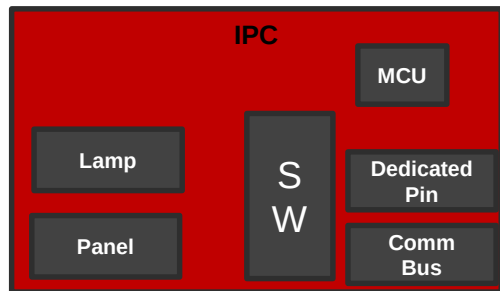
Integration Tests

Input: Integration Test Design, Unit Test Results
Output: Integration Test Result



Typical Deliverables of the Integration Tests

- Integration tests result document
- Requirements modification requests (***Change Request***)



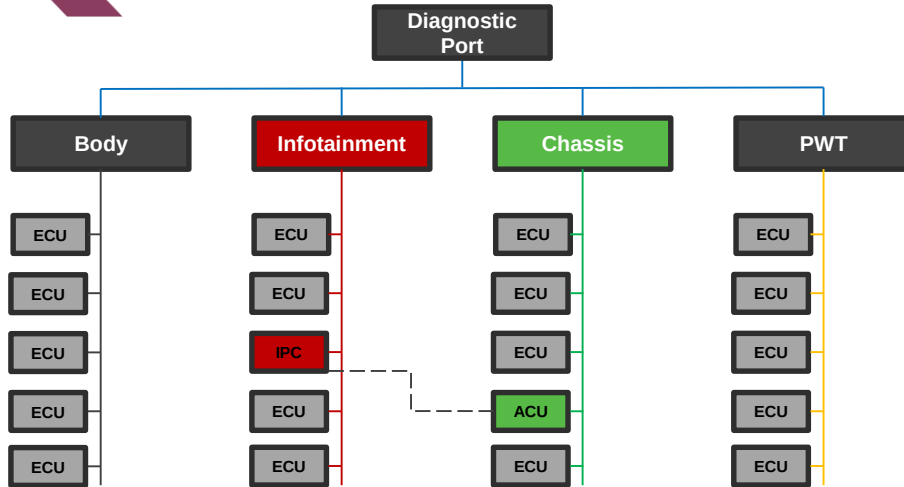


An example of a running V-cycle process

Functional Tests

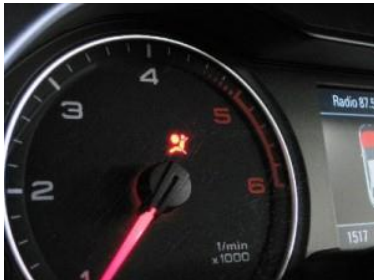
Input: Functional Test Design, Integration Test Results

Output: Acceptance Test Results



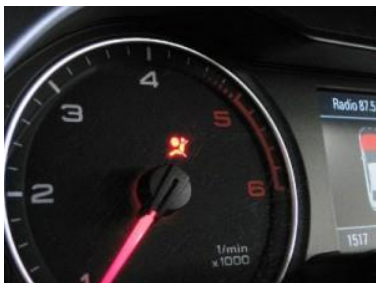
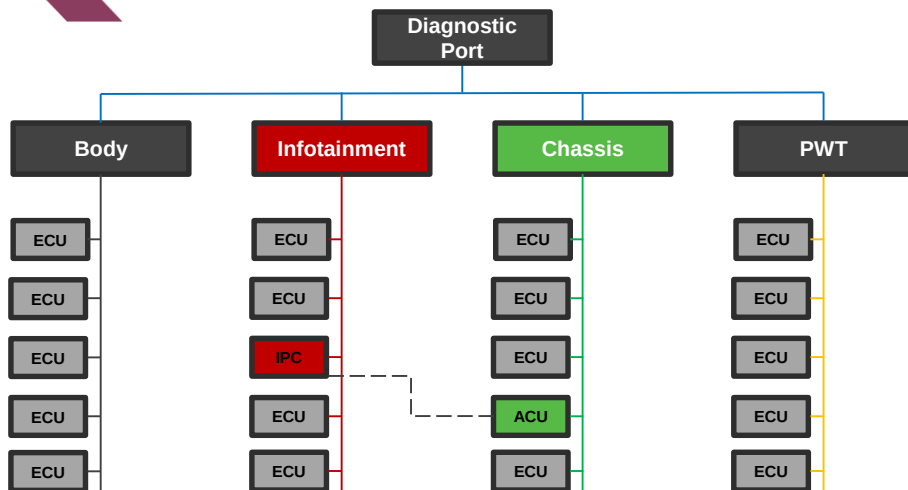
Does the system react to airbag malfunction as required (also in term of performance)?

Does the warning satisfy the constraints?





An example of a running V-cycle process



Typical Deliverables of the Functional Tests

- Integration tests result document
- Requirements modification requests (***Change Request***)

Functional Tests

Input: Functional Test Design, Integration Test Results

Output: Acceptance Test Results



Any Question?

Thank You!